Sensor de distância média

Sensor: SHARP GP2Y0A21YK0F

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Instrumentação e Medição 2024/2025

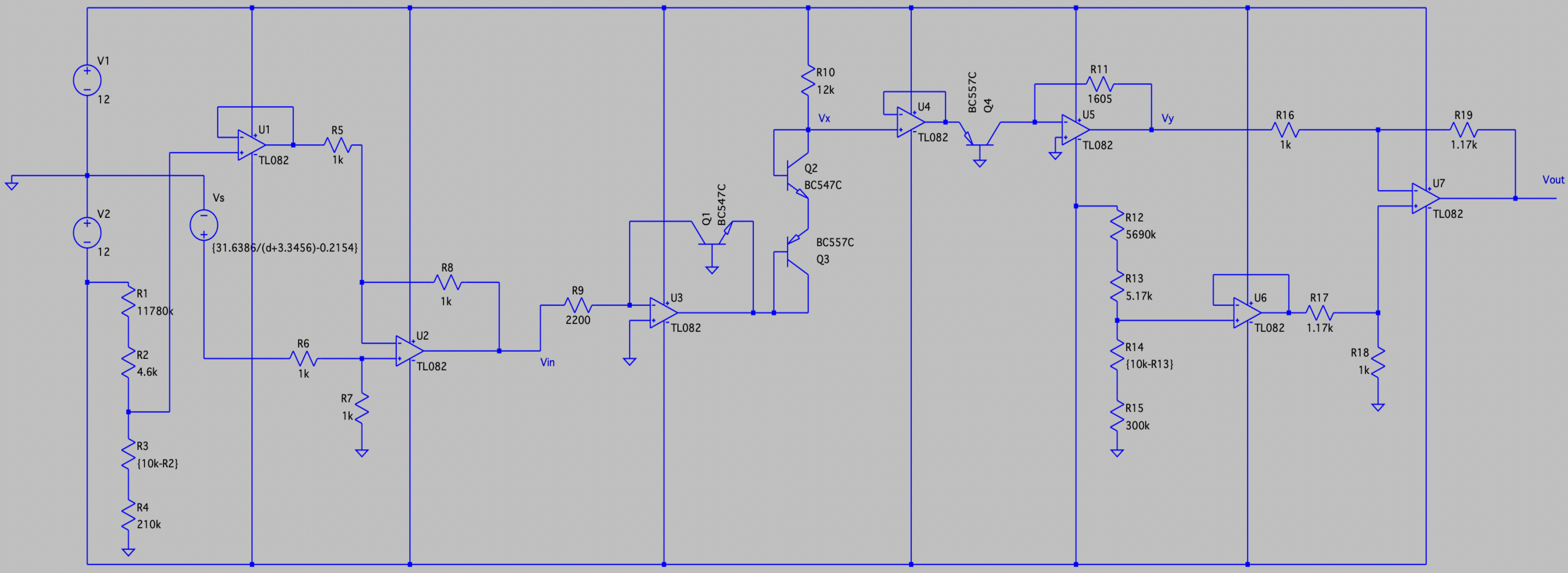
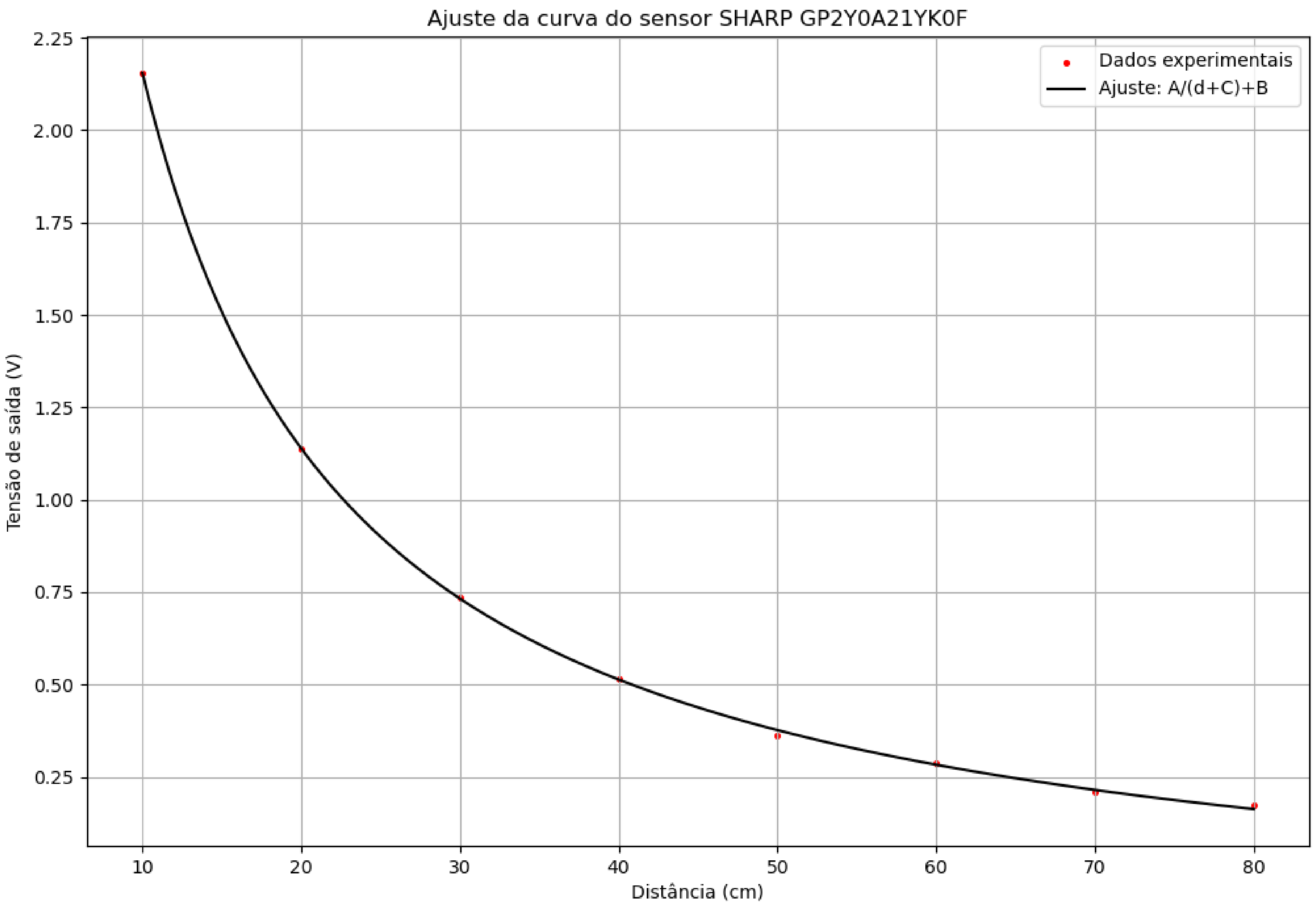

V1
12
V2
12
Vs
{31.6386/(d+3.3456)-0.2154}
R1
11780k
R2
4.6k
R3
{10k-R2}
R4
210k
U1
TL082
R5
1k
R6
1k
R7
1k
R8
1k
U2
TL082
Vin
R9
2200
U3
TL082
Q1
BC547C
Q2
BC547C
BC557C
Q3
R10
12k
Vx
U4
TL082
BC557C
Q4
R11
1605
U5
TL082
Vy
R12
5690k
R13
5.17k
R14
{10k-R13}
R15
300k
U6
TL082
R16
1k
R17
1.17k
R18
1k
R19
1.17k
U7
TL082
Vout

Diagrama de Blocos

Offset 1 Log Anti-log Offset + Ganho

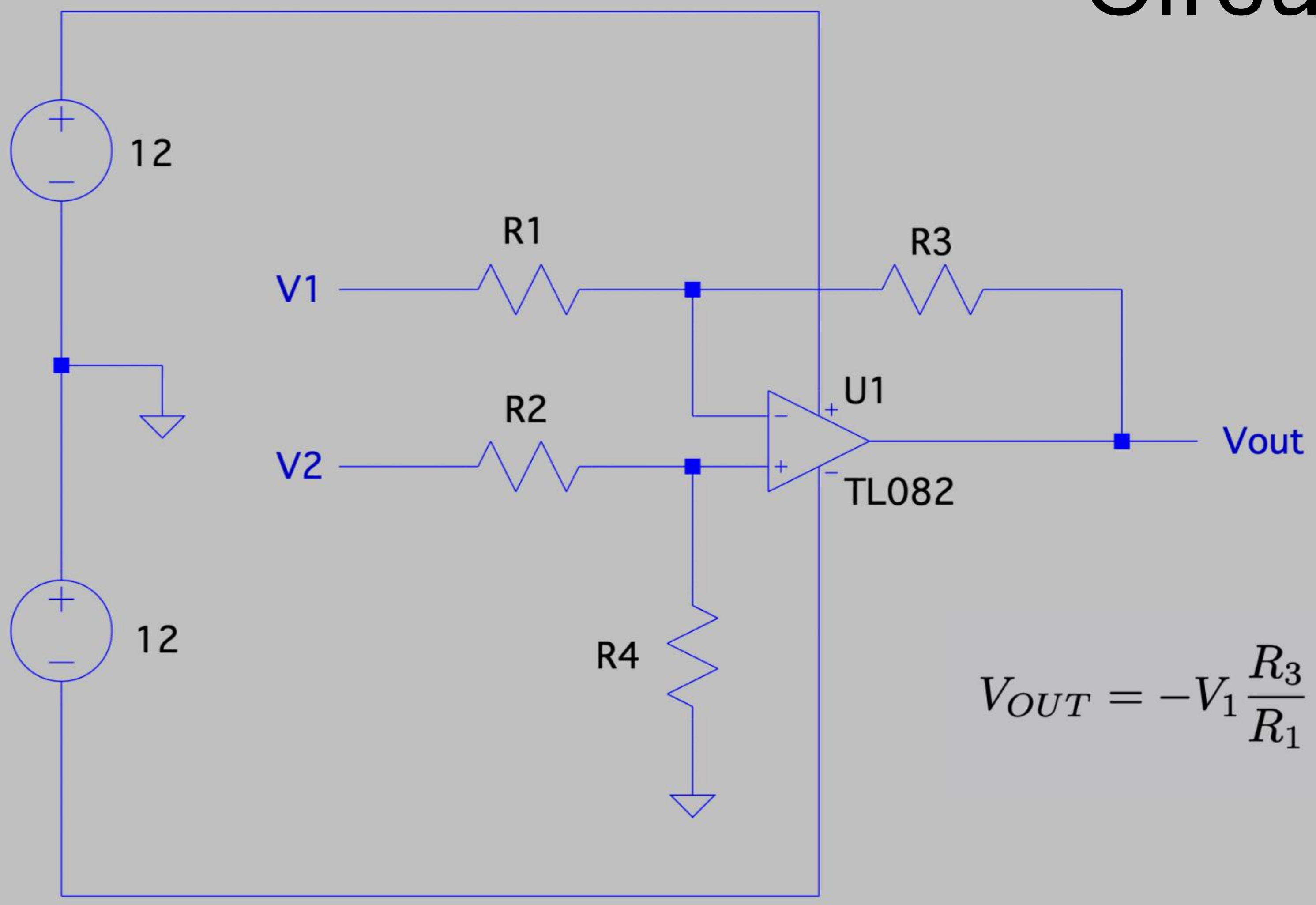
$$\boxed{\frac{A}{d+C}+B} \rightarrow \boxed{\frac{A}{d+C}} \rightarrow \boxed{-V_T \cdot ln\left(\frac{V_{IN}}{R_9 I_{S_1}}\right)} \rightarrow \boxed{V_T \; ln\left(\frac{R_9 I_{S_1}^2 I_{S_2}}{I_{REF}^2} \cdot \frac{1}{V_{IN}}\right)} \rightarrow \boxed{\frac{d}{10}}$$

$$V_{IN} = \frac{A}{d+C}$$

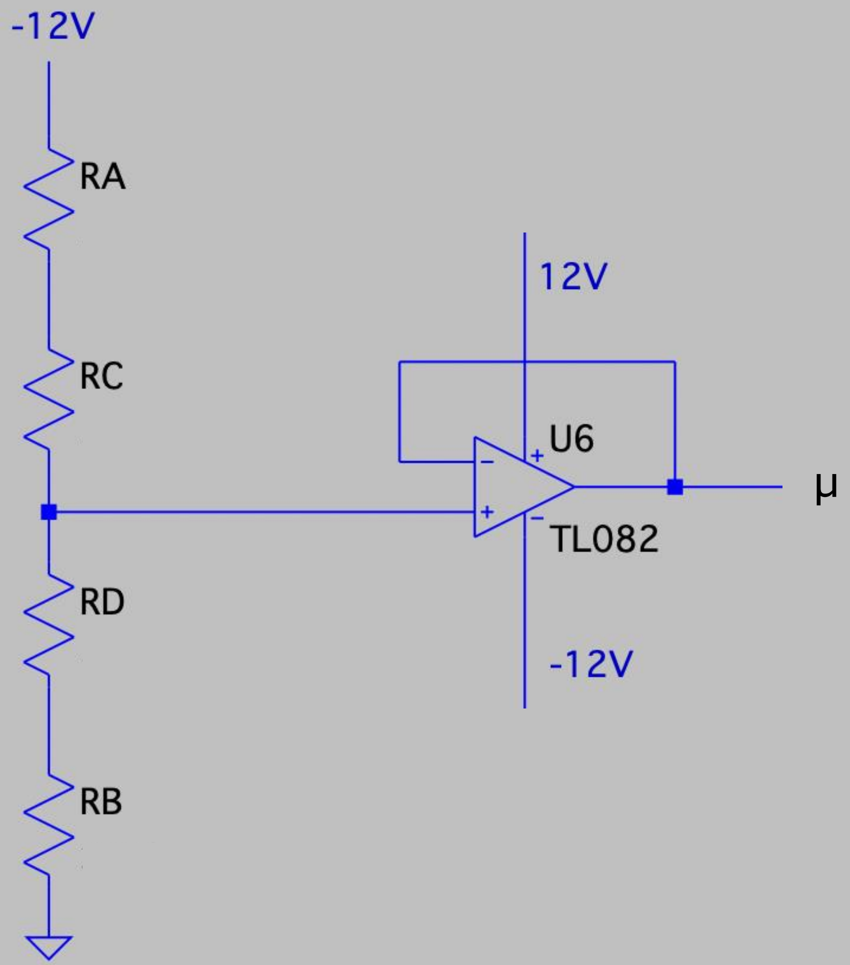


Ajuste da curva do sensor SHARP GP2Y0A21YK0F
Dados experimentais
Ajuste: A/(d+C)+B
Tensão de saída (V)
Distância (cm)
2.25
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
10
20
30
40
50
60
70
80

Circuito subtrator



$$V_{OUT} = -V_1\frac{R_3}{R_1} + V_2\left(\frac{R_4}{R_2 + R_4}\right)\left(\frac{R_1 + R_3}{R_1}\right)$$

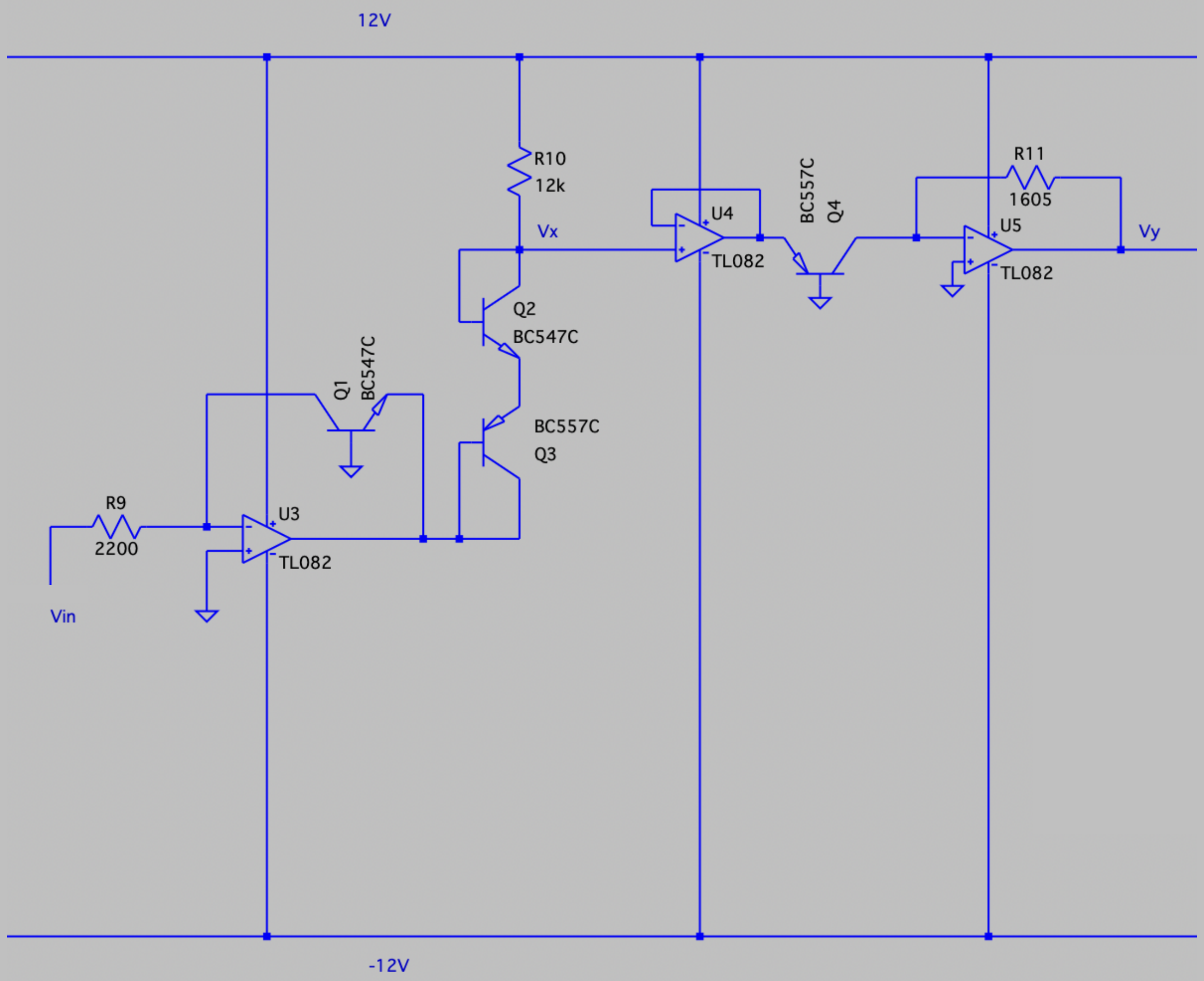


$$R_A = \frac{10k}{2}\left(\frac{12+\mu}{\Delta} - 1\right)$$

$$R_B = \frac{10k}{2}\left(-\frac{\mu}{\Delta} - 1\right)$$

$$R_D = 10k - R_C$$

Log e Anti-log



$$V_{BE_1} = V_T \cdot ln\left(\frac{I_{C_1}}{I_{S_1}}\right),\ I_{C_1} = \frac{V_{IN}}{R_9}$$

$$V_{BE_1} = V_T \cdot ln\left(\frac{V_{IN}}{R_9 I_{S_1}}\right)$$

$$V_{E_1} = V_{B_1} - V_{BE_1} = 0 - V_T \cdot ln\left(\frac{V_{IN}}{R_9 I_{S_1}}\right) = -V_T \cdot ln\left(\frac{V_{IN}}{R_9 I_{S_1}}\right)$$

$$V_{BE_2} = V_T \cdot ln\left(\frac{I_{REF}}{I_{S_1}}\right)$$

$$V_{BE_3} = V_T \cdot ln\left(\frac{I_{REF}}{I_{S_2}}\right)$$

tomando,

$$V_{IN} = \frac{A}{x},\ x = d + C$$

em que d é a distância e $V_{SENSOR} = V_{IN} + B$

$$V_X = V_{E_1} + V_{E_3} + V_{E_2} = V_T\left(-ln\left(\frac{A}{xR_9 I_{S_1}}\right) + ln\left(\frac{I_{REF}}{I_{S_2}}\right) + ln\left(\frac{I_{REF}}{I_{S_1}}\right)\right)$$

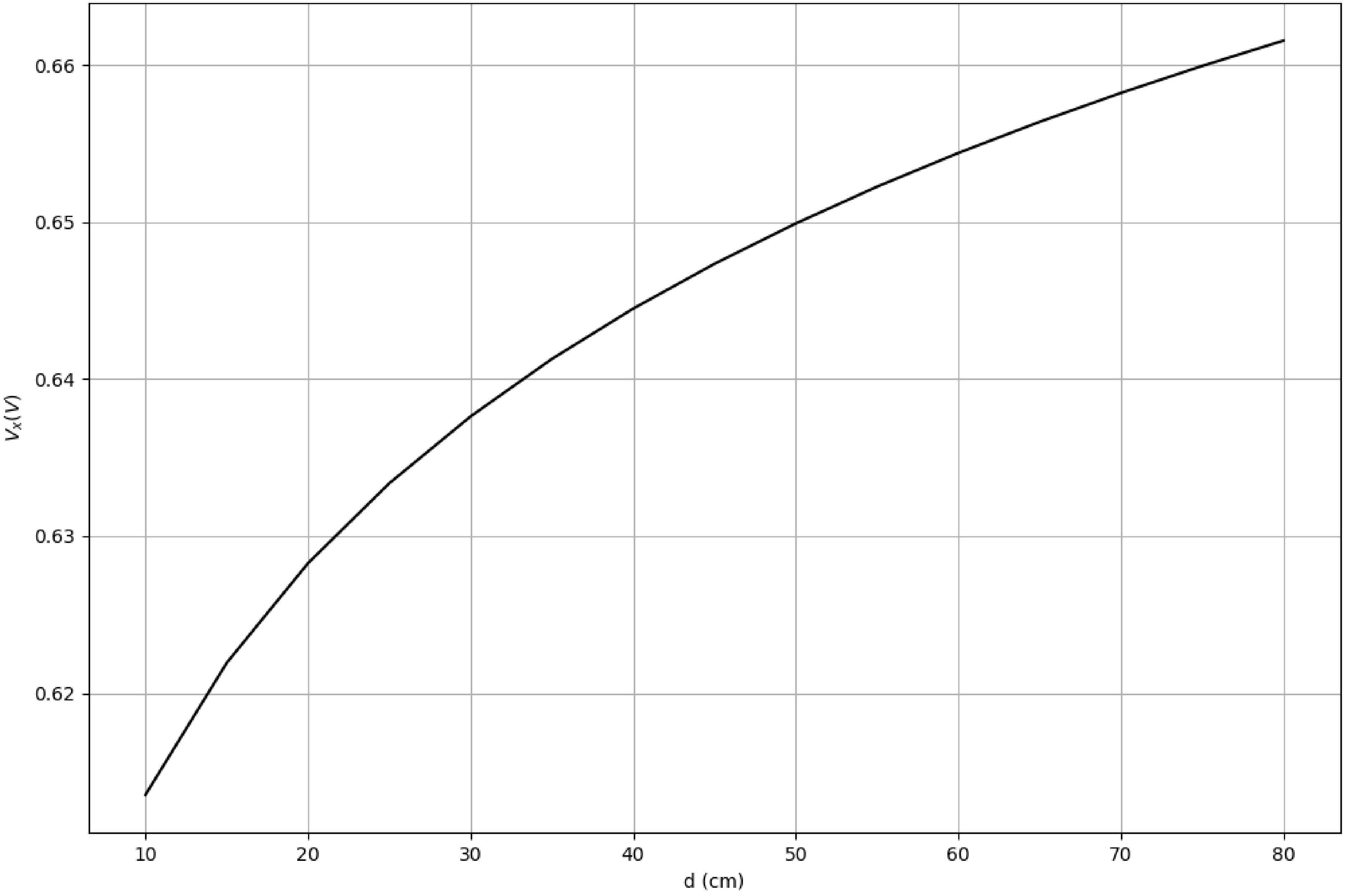
$$I_{C_2} = I_{S_2} e^{\frac{V_{EB_4}}{V_T}}$$

$$I_{C_2} = I_{S_2}\ exp\left(\frac{V_T\left(-ln\left(\frac{A}{xR_9 I_{S_1}}\right) + ln\left(\frac{I_{REF}}{I_{S_2}}\right) + ln\left(\frac{I_{REF}}{I_{S_1}}\right)\right)}{V_T}\right)$$

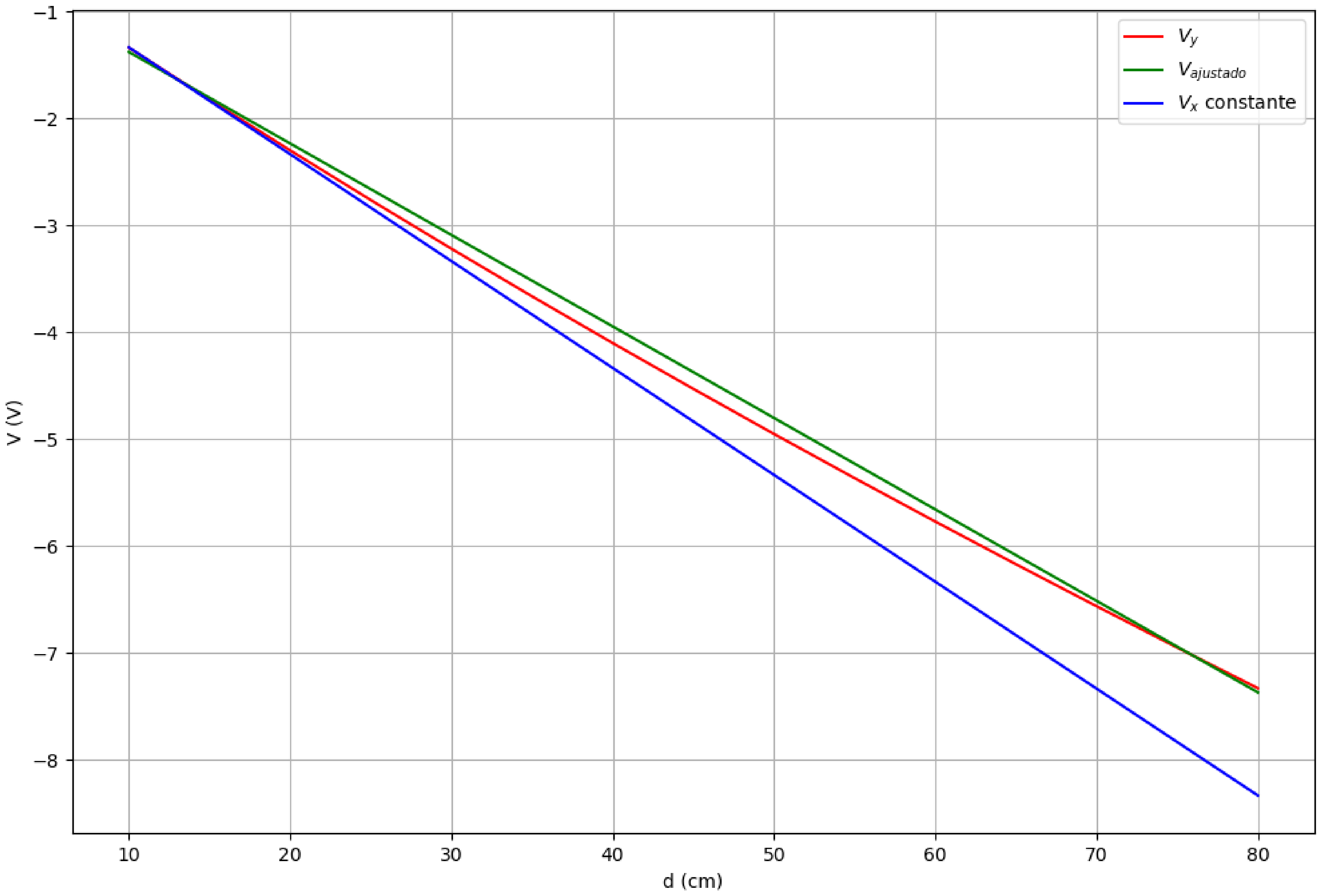
$$I_{C_2} = \frac{R_9 I_{REF}^2}{A} x,\ I_{REF} = \frac{12 - V_X}{R_{10}}$$

$$I_{C_2} = \frac{R_9 (12 - V_X)^2}{A R_{10}^2} x$$

$$V_Y = -I_{C_2} R_{11} = -\frac{R_9 R_{11} (12 - V_X)^2}{A R_{10}^2} x$$

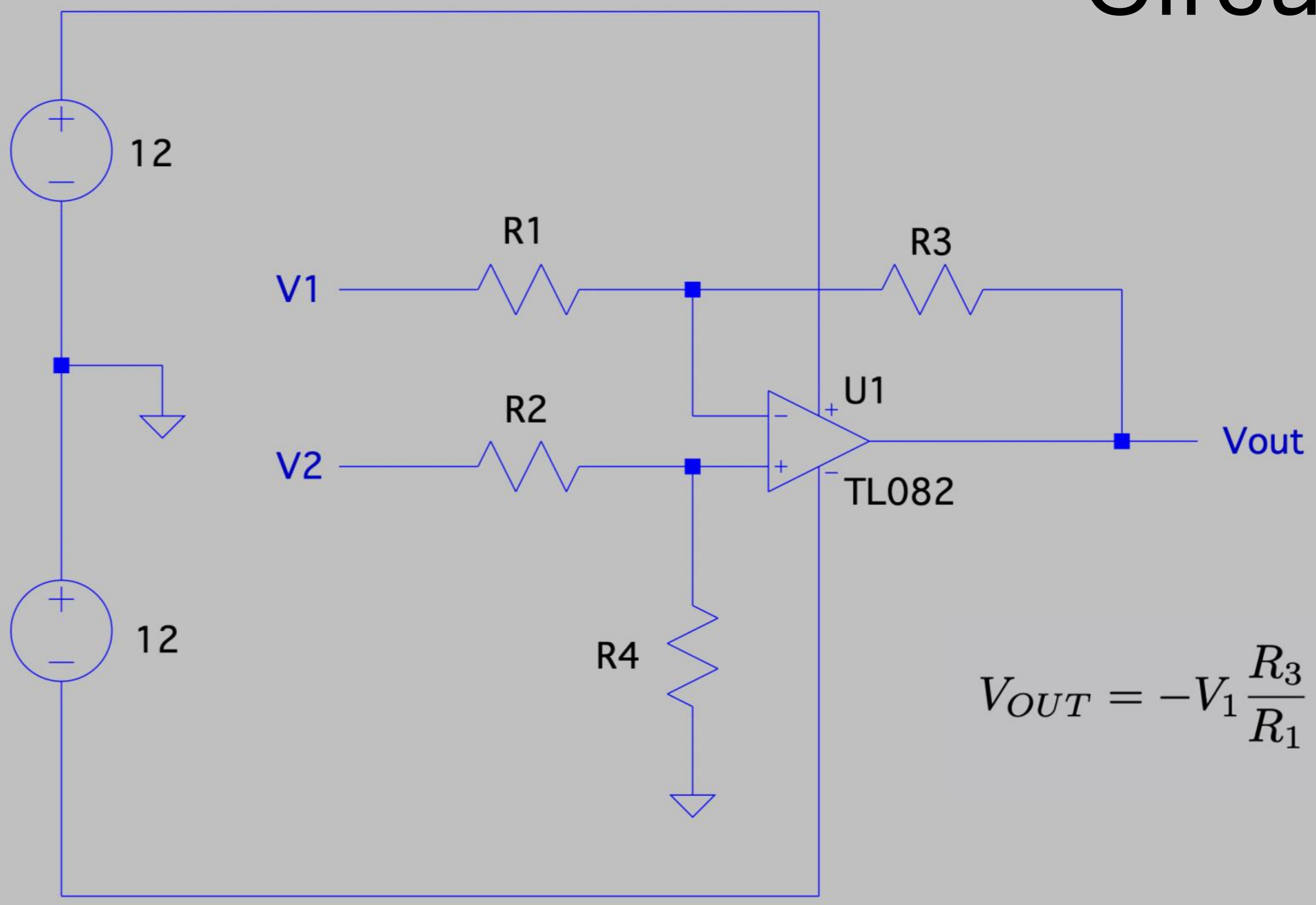


0.66
0.65
0.64
0.63
0.62
$V_x(V)$
10
20
30
40
50
60
70
80
d (cm)

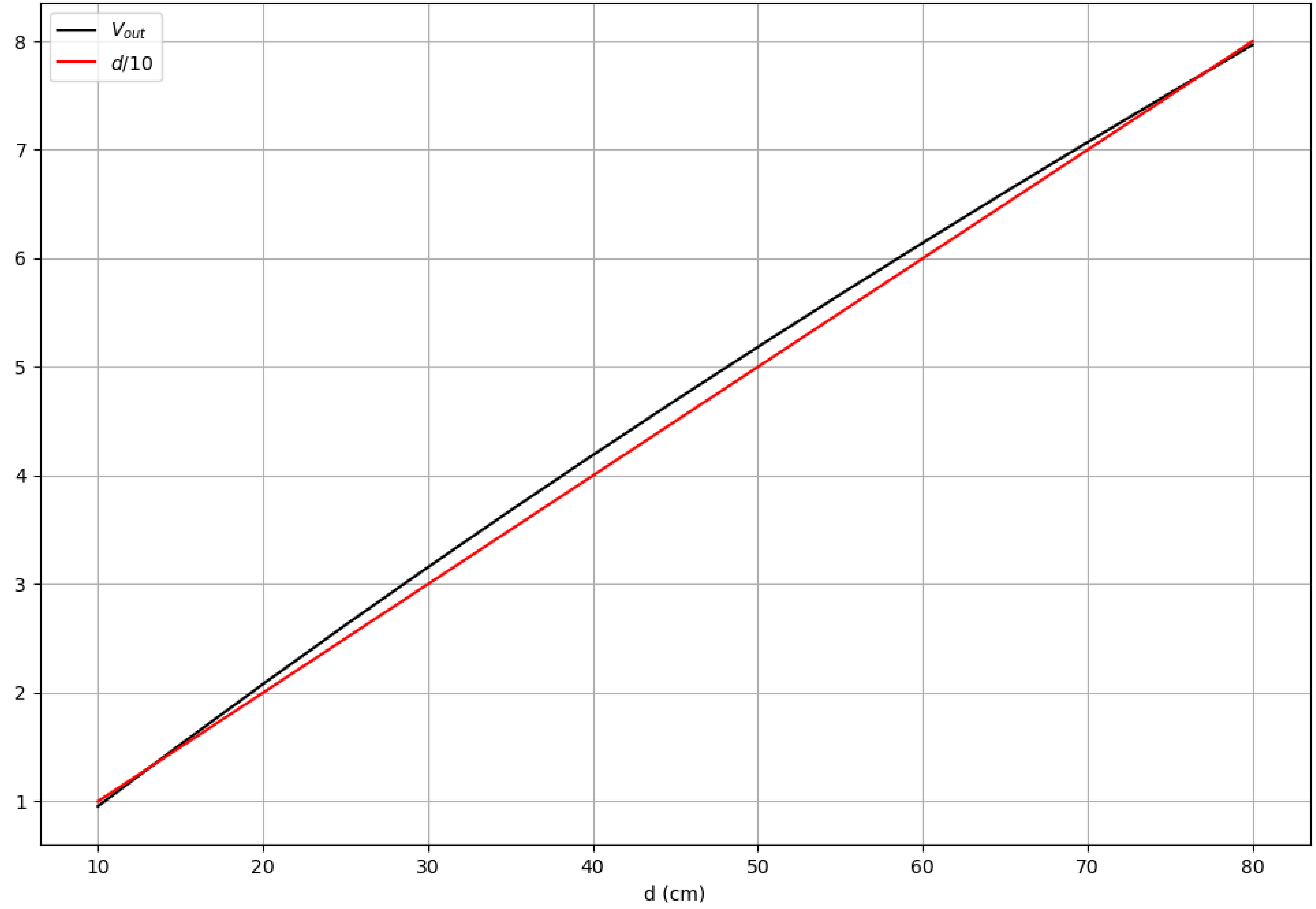


V_y
$V_{ajustado}$
V_x constante
V (V)
d (cm)
−1
−2
−3
−4
−5
−6
−7
−8
10
20
30
40
50
60
70
80

Circuito subtrator



$$V_{OUT} = -V_1\frac{R_3}{R_1} + V_2\left(\frac{R_4}{R_2+R_4}\right)\left(\frac{R_1+R_3}{R_1}\right)$$



V_{out}
$d/10$
8
7
6
5
4
3
2
1
10
20
30
40
50
60
70
80
d (cm)